

Review of Multilevel Models So Far

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1 Graph

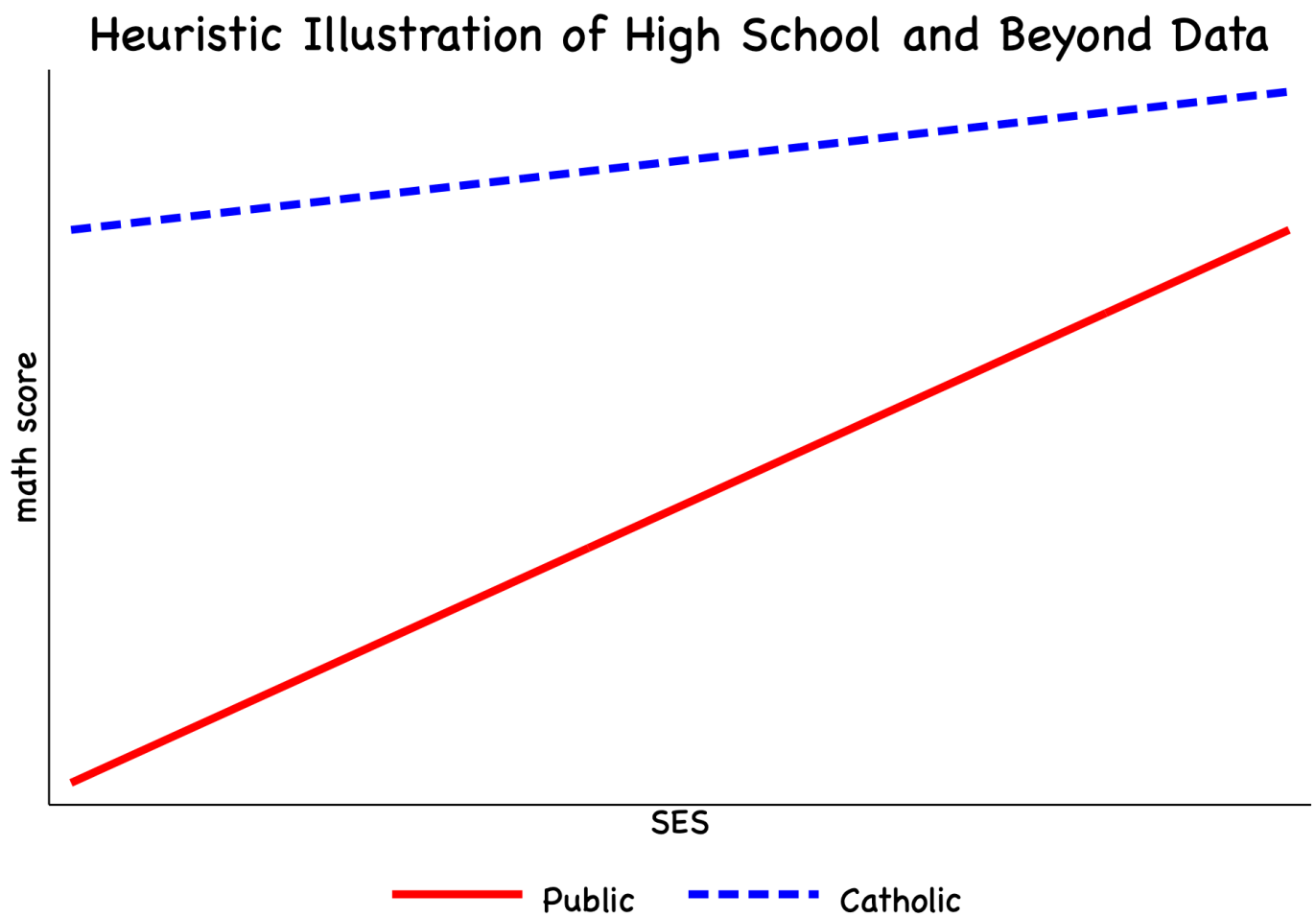


Figure 1: heuristic illustration of HBS data

2 Equation in Multiple Equation Format

$$\begin{aligned}y_{ij} &= \beta_{0j} + \beta_{1j}x + e_{ij} \\ \beta_{0j} &= \gamma_{00} + \gamma_{01}\text{sector} + u_{0j} \\ \beta_{1j} &= \gamma_{10} + \gamma_{11}\text{sector} + u_{1j}\end{aligned}\tag{1}$$

💡 Random Effects Are Optional

The random effects in Equation 1 are optional in the sense that they may or may not be included. Often only a random intercept is included. Inclusion of multiple random slopes may lead to non-convergence of the model.

3 Intra-Class Correlation Coefficient

$$\text{icc} = \frac{\text{var}(u_{0j})}{\text{var}(u_{0j}) + \text{var}(e_{ij})}\tag{2}$$

4 How Many Levels?

For the i^{th} student in the j^{th} school in the k^{th} district.

$$\begin{aligned}y_{ijk} &= \pi_{0jk} + \pi_{1jk}a + e_{ijk} \\ &\text{there are p level 1 coefficients} \\ \pi_{pj k} &= \beta_{0pk} + \beta_{1pk}r_{pj k} \\ &\text{group specific effect for the jth group} \\ &\text{there are q level 2 coefficients} \\ \beta_{pqk} &= \gamma_{0pq} + \gamma_{1pq}w + u_{pqk} \\ &\text{there are p*q level 3 coefficients}\end{aligned}\tag{3}$$

5 A Single Level Equation

Using substitution, we re-write Equation 1.

$$y_{ij} = \gamma_{00} + \gamma_{01}\text{sector} + u_{0j} + (\gamma_{10} + \gamma_{11}\text{sector} + u_{1j}) \times x + e_{ij}\tag{4}$$

We then rearrange the terms of Equation 4.

$$y_{ij} = \gamma_{00} + \gamma_{01}\text{sector} + \gamma_{10}x + \gamma_{11}\text{sector} \times x + e_{ij} + u_{0j} + u_{1j} \times x\tag{5}$$

Let's just call everything a β .

$$y_{ij} = \beta_0 + \beta_1\text{sector} + \beta_2x + \beta_3\text{sector} \times x + e_{ij} + u_{0j} + u_{1j} \times x\tag{6}$$

💡 x is ses

In the concrete example that I am using x is ses